

Revanth Palaparthi

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EDUCATION

Indian Institute of Technology, Dharwad

B.Tech in Computer Science and Engineering

Dec 2021 – July 2025

Relevant courses: Artificial Intelligence, Software Systems & Laboratory, Data Structures & Algorithms, Data Base Management Systems, Design and Analysis of Algorithms, Operating Systems, Compilers, Deep Learning, Generative AI, Agentic AI.

WORK EXPERIENCE

A Comparative Study of Deep Learning Architectures for Cervical Cell Segmentation in Pap Smear Images (IEEE INCIT 2025, IIT Dharwad Research Publication):

- Conducted a comprehensive comparative study of eight deep learning models (U-Net, U-Net++, Attention U-Net, DeepLabv3+, PAN, SegFormer, YOLOv8, YOLOv11) for cervical cell segmentation on the SIPaKMeD medical image dataset.
- Engineered a modular PyTorch-based segmentation pipeline with patch-wise image cropping (512×512), advanced data augmentation (rotation, flipping, elastic transformation), and precise mask handling for improved boundary recognition.
- Achieved top performance using U-Net++ (Dice: 0.84, IoU: 0.78) and demonstrated competitive real-time inference with YOLOv11 (Dice: 0.83), highlighting trade-offs between accuracy and speed.
- Evaluated models using clinical-grade metrics (IoU, Dice, Precision, Recall, F1-score, Accuracy), supporting data-driven architecture selection for medical image analysis and diagnostic systems.

Defence Research and Developmental Laboratory (DRDL) (Summer intern):

- Developed scalable, fault-tolerant network applications to optimize communication within a distributed system, utilizing packet retransmission strategies to ensure reliability and performance.
- Engineered real-time interaction protocols, enhancing session management and optimizing packet retransmission by up to 50%.
- Architected a simulation using TCP sockets for decentralized network management, resulting in a 50% increase in throughput.
- Deployed Ring Token Protocol to streamline communication and reduce latency across distributed nodes.

PROJECTS

Smart Email Knowledge Assistant:

- Implemented an intelligent email assistant integrating Gmail, Postgresql and ChromaDB for capturing, storing and retrieving email conversations.
- Enabled contextual search and question-answering over inbox threads using Langchain and GPT-based models.
- Built a continuous email synchronization pipeline to process communications and maintain searchable repository of inbox data.
- Developed event detection capabilities to identify important events communicated through emails that may not be reflected in calendar applications.

DocMind AI - Multi-Document Knowledge Assistant:

- Developed a GenAI-powered multi-document RAG system enabling semantic search and question answering across 50+ PDF documents using vector embeddings and retrieval-augmented generation.
- Implemented a citation-aware retrieval pipeline that delivers source-grounded responses with document and page-level references for transparency and verifiability.
- Engineered advanced document intelligence features including cross-document question answering, comparative analysis, automated study-note generation, and quiz creation.

Algorithmic Trading System with Automated Portfolio Management:

- Developed an automated trading system integrating Zerodha Kite API, enabling real-time market data streaming, order execution, and trade monitoring.
- Implemented AI-driven trading strategies using technical indicators (RSI, MACD, Bollinger Bands) and historical data analysis, improving decision-making efficiency.
- Designed a backtesting module with Python and Pandas, enhancing strategy evaluation and optimizing investment performance.
- Engineered a secure authentication system with OAuth2 and multi-user access control, ensuring robust data protection and regulatory compliance.

Dynamic Social Interaction Platform:

- Engineered a basic social media platform with user authentication, posting, commenting, liking, and friending functionalities.
- Implemented server-side logic using Python, efficiently managing data transactions and server responses.
- Designed SQL schema and automated data population, streamlining database setup.
- Coordinated application flow through a modular Python architecture, enhancing user engagement and platform usability.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java

Database Management: MySQL, MongoDB, Postgresql

Web-development: HTML, CSS, JavaScript, PHP, React, Node.js, Express.js, Tailwind

Scripting: Python, Bash

Libraries & Tools: NumPy, Pandas, Git, GitHub, Linux, Latex, Beamer, Flask

ACHIEVEMENTS

- Awarded 3rd place in the Parsec 2024 Treasure Hunt**, showcasing strong problem-solving and teamwork skills in a highly competitive environment.
- Achieved **Led sponsorship acquisition for E-Summit '22 and '23**, securing over 10 industry leaders, resulting in increased event credibility and extensive networking opportunities for participants.
- Ranked Top 5 in the 'DEVHACK' event at PARSEC**, IIT Dharwad's prestigious tech fest, by designing and developing an innovative web solution, engaging 100+ participants, and fostering collaboration among peers.
- Attended the **9th International Conference on Information Technology (INCIT 2025)**, Thailand, organized by IEEE.

LANGUAGES

English, Hindi, Telugu